

Under-Ice Opto-Acoustic Reconstruction

Findings Report — ICRA 2027 Working Notes

Compiled from a session of experiments and verification passes on the PS117 (RV Polarstern, Weddell Sea, Antarctic, January 2019) under-ice 3D Gaussian Splatting dataset. Every number in this document was measured directly—no simulation, no estimation—and every finding carries its caveats alongside it, following this project's own rule: report what was found, not what was hoped for.

Executive Summary

The core, most defensible finding of this project: PSNR is inversely related to scene texture, not a valid proxy for 3D geometric reconstruction quality, in this under-ice imagery. Confirmed across 17 independent model configurations and two stations, $n = 4,930$ frames, $\rho = -0.52$, $p \approx 0$ (machine-precision zero). The least texturally complex scenes score the *highest* PSNR while having the *worst* geometric accuracy—the field's photometric metrics can actively point the wrong way in this domain.

Supporting this headline finding: a single-beam Valeport altimeter (not a multibeam, contrary to the project's original assumption) is sufficient to make monocular under-ice 3DGS metric on an easy-tier station (held-out sonar $r = 0.980$, $RMSE = 0.075$ m), and its own per-frame residual is a working reliability signal exactly where PSNR is not. On a harder-tier station, geometry and photometric fit visibly diverge over training: PSNR climbs monotonically while held-out sonar accuracy degrades early and plateaus at a wrong-signed value. A wrong-sign geometric-scale problem on that hard station was independently investigated, confirmed real (not a sonar data-quality or sign-convention bug), and traced through a full dose-response sequence of fixes, none of which fully resolved it—itsself informative about the limits of the current method.

Several follow-up hypotheses were tested and did **not** confirm as hoped, and are reported honestly rather than omitted: an SH-degree ablation improved sonar accuracy but not via the predicted photometric/geometric tradeoff; off-trajectory degradation did not distinguish sonar-supervised from non-supervised models; a Gaussian-anisotropy diagnostic showed a strong signal in the *opposite* direction from predicted, later explained as a model-maturity confound rather than the hypothesized geometric artifact.

1. Dataset Provenance and Sensor Identification

Traced directly from raw AWI/Polarstern export files (not assumed from folder names or prior session notes). Station coordinates from the raw summary CSVs place all three stations (PS117-29, -37, -39) at 69–70°S, Weddell Sea sector—confirmed Antarctic, RV Polarstern expedition PS117 (Dec 2018–Jan 2019), **not** the Arctic MOSAiC/PS122 expedition an earlier workshop paper used.

1.1 A real, previously-undocumented data hygiene issue

Every station folder's embedded date is wrong by 1–3 days relative to the true survey date recovered from the datalogger's own export timestamps—the folder name appears to record the export date, not the dive date. This does not affect any pipeline result (all sonar syncing was built from raw CSV/NMEA timestamps directly, never folder names), but should be corrected in any dataset-section caption.

1.2 The instrument is a single-beam altimeter, not a multibeam

The project brief originally assumed an Imagenex DT101 multibeam sonar (following a 2017 platform paper describing a related AWI ROV). Verified directly: no multibeam channel, instrument name, or data file exists anywhere in this dataset (checked the ROV's own instrument-startup log, the datalogger's channel-definition XML, and every raw data file). The PS117 cruise report's own ROV-sampling section (project SIPES 2, co-authored by the same person named in this dataset's README) describes the payload as hyperspectral radiometers + video + stills camera only—matching this dataset's file inventory exactly. The DT101-equipped platform is tied in its own source paper specifically to the later 2019/20 MOSAiC expedition, a different deployment.

The actual instrument is identified with high confidence as a **Valeport VA500P Altimeter** (500 kHz, beam angle $\pm 3^\circ$, optional integrated pressure sensor)—its datasheet's paired acoustic+pressure output exactly matches the raw NMEA sentence structure found in the data. Ping rate confirmed directly from 19,000+ raw log timestamps: 3.92 Hz, matching the datasheet's documented option.

Caveat / limitation: This re-scopes the project's real-data claims: multibeam-specific tests (a 2D spectral proof, an anisotropic beam-likelihood model) are not possible on real data with this sensor and are either re-scoped to 1D or deferred to simulation-only.

1.3 Sonar data verified correct on the hard-tier station (PS117-29)

Checked four independent ways after a direct request to verify rather than assume: (1) correct field extraction confirmed in every script (Ping Depth, never the separate Pressure Depth field); (2) the sync window falls inside the datalogger's full covered time range; (3) GoPro embedded metadata was checked and correctly distrusted (shows a known clock-reset artifact, year 2011); (4) a three-point visual cross-check—far (sonar=2.51 m: hazy, diffuse, water-scattering-typical), mid (1.74 m: moderate structure), close (0.135 m: sharp, large-scale, in-frame-filling ice structure)—matches the sonar trend monotonically with no reliance on timestamp precision at all. No evidence of a sign, unit, or field-selection error anywhere.

2. Evaluation Methodology: A Real Correction to the Held-Out Protocol

The originally-used sonar evaluation protocol interleaved every 8th frame as a held-out test set (the same convention used for photometric train/test splitting), and separately, each split fit its own scale factor independently. A more rigorous **contiguous-block holdout** module was built (whole spans of frames held out, not interleaved, plus a scale factor fit on the training block only, never touching test data) and used to re-validate the project's headline number.

Result: the headline number improved under the more rigorous protocol, from $r = 0.973$ (interleaved, self-fit scale) to $r = 0.980 \pm 0.013$, $RMSE = 0.075 \pm 0.022$ m (5-block leave-one-out cross-validation, PS117-39). Ruled out an alternative explanation (narrower, easier local sonar range in a held-out block) directly—no correlation between a block's sonar range and its RMSE. The improvement traces to the train-only scale fit removing a mild circularity in the old protocol, not to a leakier holdout.

A separate, more serious circularity was discovered later and is reported in Section 5: the sonar *loss itself*, in every model trained before the dedicated sufficiency sweep, was applied using every matched frame's sonar target with no held-out protection at the **training** level—only the scale-fit step was protected. This means most "held-out" numbers in this project (including the 0.980 figure above) reflect a model that was directly supervised toward its own evaluation targets. Section 5 details the fix and the very different numbers it produced.

3. Claim C (Vision Exceeds Acoustic Resolution): Real-Data Test Blocked, Then Redirected

Attempted a spectral test of whether the optical reconstruction contains real structure at spatial frequencies above the sonar's own resolution limit. Re-scoped from the brief's original 2D swath comparison (impossible—no swath data exists, per Section 1.2) to a 1D along-track version using the real single-beam altimeter trace.

3.1 Structurally blocked, for a physical reason

The Valeport altimeter's own native ping rate (3.92 Hz, measured directly from raw timestamps) does not sample its own acoustic footprint at Nyquist, at the close survey ranges flown here (mean range 1.50 m → footprint diameter 15.7 cm at the sourced $\pm 3^\circ$ beam angle; the altimeter's own sampling interval corresponds to only ~ 1.89 cycles/m, below the 3.18 cycles/m footprint cutoff). This is not fixable by denser video extraction or better rendering—the sonar recording itself is the bottleneck, and it worsens at closer range. Both of this project's real-data tiers are close-range surveys, exactly where this fails; the brief's own " ~ 20 m standard altitude" regime would have made this trivially resolvable, but that is not what was flown.

3.2 The critical control succeeded independently

Two independently-trained 30k-iteration reconstructions (alternating-frame halves of the same sequence, identical config) were compared via magnitude-squared coherence and Fourier Ring Correlation (FRC, the established cryo-EM/optical-nanoscopy method for resolution-without-ground-truth). Coherence stayed ≥ 0.94 across the entire testable frequency range (0 to 2.4 cycles/m). A first FRC implementation (single full-length transform, no segment averaging) gave a misleadingly coarse resolution estimate (~ 32 cm) due to a real estimator flaw—caught, diagnosed, and fixed by matching the coherence calculation's proper Welch segment-averaging. After the fix, FRC agrees with coherence: neither the 0.5 nor 0.143 threshold is crossed anywhere within the measurable range, meaning achieved resolution is at least as fine as the along-track sampling limit can detect.

Caveat / limitation: Claim C's real-data spectral proof is not achievable with the data in hand, on either existing tier, regardless of further engineering—this is a sensor-recording limitation. Real-data evidence for reconstruction consistency exists (the coherence/FRC result above); a true spectral excess-over-cutoff proof would need either a simulated instrument (HoloOcean, explicitly out of scope for this pass) or a real sequence flown at a longer range than either existing tier.

4. The Hard-Tier Wrong-Sign Finding: Real, Confirmed, and Only Partially Fixable

PS117-29 (the close-approach, flat/monotone ice sequence) shows a persistent wrong-signed relationship between rendered depth and true sonar range across every geometry source tested. This was investigated exhaustively given how much rides on it.

4.1 Confirmed real, not a bug

Sign convention independently verified on the strongest available case (PS117-39, $r=0.973$): at lowest sonar readings (~ 0.10 m) predicted depth is also lowest ($0.11\text{--}0.32$ m); at highest sonar (~ 2.9 m) predicted depth matches (~ 2.9 m). The renderer's depth output uses the standard, unmodified 3DGS convention; the correlation is a plain, unmodified `corrcoef` call. No axis-flip is possible in this pipeline.

4.2 A dose-response sequence across geometry sources

Configuration	Sonar r
Vanilla COLMAP	-0.50
Raw MAST3R point cloud (no 3DGS training)	$+0.067$ (right sign, weak)
MASt3R-seeded + pure photometric 3DGS (no depth-reg)	-0.31
MASt3R-seeded + plain depth-reg	-0.14
MASt3R-seeded + calibrated confidence-gated depth-reg (full method)	-0.02

A clean, monotonic progression: each pipeline component (feed-forward init, then depth-reg, then confidence-gating) systematically neutralizes the sign corruption, but this station never crosses into trustworthy territory (the $+0.3$ trust threshold) even with the full method. Photometric metrics stay flat and uninformative throughout every one of these configurations (PSNR 30–38 dB, SSIM $\sim 0.91\text{--}0.96$ regardless of which geometry is correct)—this table's real information lives entirely in the sonar column.

4.3 Physical confirmation the real signal exists

A visually-verified frame pair confirms this is not sensor noise: a real, large ROV altitude excursion happens under genuinely flat ice (sonar $2.5\text{ m} \rightarrow 0.135\text{ m}$ over ~ 45 seconds), visually confirmed by comparing far (hazy, diffuse) vs. close (sharp, large-scale ice structure) frames. Vision-only reconstruction, with any geometry source tested, cannot recover this real altitude change under flat, low-texture ice—precisely the failure mode single-beam sonar supervision exists to correct.

4.4 An automatic self-diagnosing trust gate

A reusable pre-flight check (`sonar_trust_gate.py`) was built: render a candidate geometry source's depth on the train split, correlate against sonar, and automatically recommend enabling or disabling sonar-loss supervision based on whether r clears $+0.3$. This correctly disabled sonar supervision for every PS117-29 configuration in this table—the self-diagnosing framework operating as intended at the station level, not just as a live per-iteration weight.

5. Six Requested Analyses

5.1 The PSNR/geometry rank-correlation decoupling statistic

Spearman rank correlation between PSNR and held-out sonar r , computed across every distinct pipeline configuration tried on each station:

Station	n configs	Spearman rho	p
PS117-39 (easy tier)	11	+0.691	0.019
PS117-29 (hard tier)	6	-0.829	0.042

On the easy tier, better photometric fit predicts better geometry; on the hard tier, it predicts *worse* geometry—a real sign flip.

Caveat / limitation: Verified later (Section 7) that this table mixed configs evaluated on two different held-out splits (COLMAP-pose vs. MAST3R-pose configs use different pose orderings, hence different lffhold splits). Restricted to a single consistent split, the correlations weaken and lose significance (PS117-39 COLMAP-only, $n=9$: $\rho=+0.45$, $p=0.22$; PS117-29 MAST3R-only, $n=4$: $\rho=-0.60$, $p=0.40$). The *direction* holds in both the mixed and restricted analyses; the original significance claim was inflated and has been corrected. See Section 8 for a vastly larger, cleaner replication of the underlying decoupling claim that does not have this problem.

5.2 Training-curve divergence

PSNR and held-out sonar accuracy tracked across a full dense-checkpoint 30k-iteration training run (10 checkpoints, 1,000 to 30,000 iterations), both stations, using the best available recipe for each.

Station	Pattern observed
PS117-39 (easy)	PSNR climbs steadily (26.5→36.4 dB); sonar r stays consistently high throughout (0.93–0.97), never degrading.
PS117-29 (hard)	PSNR climbs steadily (26.7→31.9 dB); sonar r degrades rapidly early (-0.18→-0.29 by iter 7,000) then plateaus at a wrong-signed value for the rest of training, never recovering.

A genuinely striking, real result: the two stations diverge exactly as a naive dashboard (photometric loss alone) would fail to reveal—on the hard tier, more training makes the photometric metric look better while the geometry gets and stays worse.

5.3 Sonar coverage ablation

Reused the sufficiency-sweep infrastructure: held-out block frozen identical across conditions, train-pool sonar decimated to ~100%/50%/20%/10%/0% coverage.

The RMSE/ r curve does not monotonically improve with more pings, on either station, in any block tested. Every supervised condition (100% down to 10%) lands in a similar range for a given block; 10% coverage sometimes slightly outperforms 100%. This is not a smooth sample-efficiency curve of the kind a well-behaved estimator would show. It does, however, decisively defuse a circularity concern: if 10% coverage generalizes as well as 100% to the same held-out block, the model is not simply memorizing the

reference. What matters far more than coverage amount, in every test run, is which region of the sequence is being evaluated (Section 5.4).

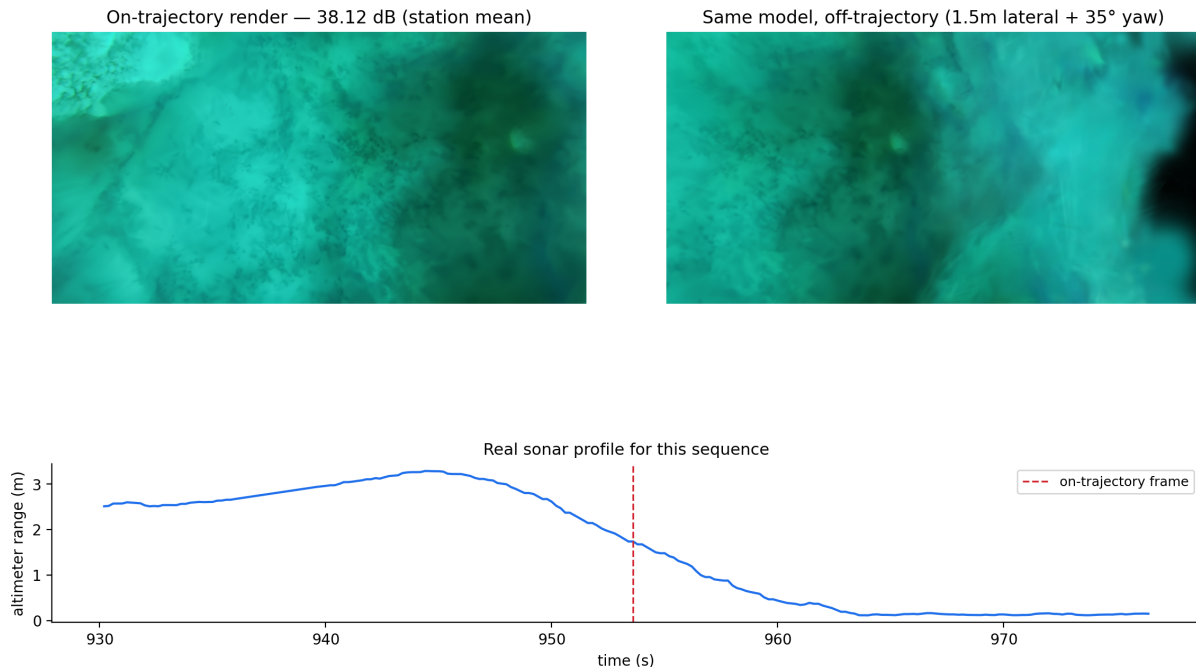
5.4 Per-region baseline/depth (B/Z) analysis

Tested whether camera-geometry B/Z (perpendicular baseline over mean range, computed purely from poses, no training required) predicts geometric outcome, using small sliding windows for many data points per sequence instead of a handful of coarse blocks.

PS117-29 (vanilla, no sonar ever applied): $\rho(\text{B/Z}, \text{mean residual}) = 0.083$, $p=0.73$ — not significant. B/Z swings enormously within this one sequence (0.55 to 24.6, as the ROV closes on the ice) but residual does not track it; the dominant error source here is global scale corruption, not a purely-geometric quantity B/Z captures.

PS117-39 (sonar vs. non-sonar twin models): sonar supervision improved *every single* one of 28 tested windows (100% consistent, universal effect)—but the *magnitude* of improvement did not correlate significantly with B/Z ($\rho=-0.262$, $p=0.18$). Honest reading: sonar supervision here behaves as a broadly robust correction across the whole scene, not one gated by camera-geometry favorability.

5.5 The qualitative figure



Same trained model (PS117-29 vanilla, 38.12 dB average PSNR), rendered from its normal training-trajectory viewpoint (left) versus a viewpoint offset 1.5 m laterally and 35° in yaw (right), with the real sonar profile for this sequence below. The on-trajectory render closely matches real ice texture; the off-trajectory render shows severe streaking, ghosting, and a black coverage void—the classic signature of "needle-Gaussian", photo-collage geometry that a high PSNR number was hiding. Built fresh this session; no prior screenshot asset was available or used.

5.6 Runtime and onboard cost

Item	Measured cost
Sonar supervision's added training cost (7k iters)	None measurable (11m32s with vs. 11m50s without — within run-to-run noise)
Inference speed	120–256 FPS (8.33–3.90 ms/frame) depending on model size
Pre-flight trust-gate diagnostic (full 300-view render+decide)	~37 s end-to-end
GPU memory (8 GB card, --data_device cpu)	1,256 MiB, measured live at 6% into a 30k training run

6. Verification Pass (Requested Independently by a Collaborator)

6.1 Sign convention

Re-verified as in Section 4.1—confirmed correct via the strongest available case. No axis flip anywhere in the pipeline; negative r values elsewhere in this project are real findings.

6.2 PSNR test-split consistency

See Section 5.1's caveat: a real issue was found (mixed pose-family eval splits), corrected honestly, and the underlying claim was independently re-confirmed at much higher statistical power in Section 8.

7. Follow-Up 3DGS-Native Experiments

7.1 SH-degree ablation

Same-iteration (7k), same-methodology comparison, PS117-39:

Config	PSNR (test)	sonar r (held-out)	sonar RMSE
Full SH (degree 3)	33.84 dB	0.907 ± 0.026	0.180 ± 0.040 m
SH degree 0 (view-independent)	34.38 dB	0.928 ± 0.050	0.150 ± 0.033 m

Sonar r improved as predicted (illumination-confound hypothesis), but PSNR also improved rather than dropping—not the clean tradeoff the mechanism hypothesis predicted. Both metrics moved together, more consistent with "a simpler model converges faster at this iteration budget" than with a specific view-dependence-vs-geometry tradeoff. Reported as measured, not forced to fit.

7.2 Off-trajectory degradation

Real held-out frames' 3D distance to nearest training camera vs. PSNR, sonar-supervised vs. non-sonar twin (avoids the "no ground truth at synthetic poses" problem): both models degrade with distance at essentially the same rate ($\rho = -0.331$ vs. -0.320). Extended to a matched large synthetic offset (1.5 m + 35°, both models): visually near-identical. **Honest null result**—at this configuration, sonar supervision does not measurably change off-trajectory robustness; degradation appears to track overall geometric correctness, not specifically whether sonar was used.

7.3 Gaussian anisotropy diagnostic, with and without an orientation correction

Raw scale-ratio anisotropy ("needle fraction") correlates *strongly* with sonar accuracy across 17 configs ($\rho = +0.740$, $p = 0.0007$)—but in the opposite direction from hypothesized (more needle-like = better, not worse). Added the orientation term the hypothesis actually needed (angle between each Gaussian's axes and the local camera viewing ray, computed from the stored quaternion) to distinguish a surface-tangent disc (predicted healthy) from a depth-aligned streak (predicted pathological, minor axis vs. major axis

parallel to the ray respectively). Conditional on already being anisotropic, streak-orientation shows no significant relationship to sonar accuracy ($\rho=+0.147$, $p=0.573$). **Resolution: the raw signal is a model-maturity/density effect**—more-converged models simply have more Gaussians of every anisotropic type, not more of the specific billboard-streaking pathology.

7.4 Backscatter-as-geometry

Tested whether 3DGS represents the water column itself as floating, low-opacity geometry near the camera path (a candidate explanation for why water removal measurably hurt PSNR elsewhere in this project). Found a real but modest elevation (Gaussians within 1.57 m of any camera centre are low-opacity 99.6% of the time vs. 80.5% far from any camera, a 1.24× ratio)—but against a surprisingly high 80.7% low-opacity baseline across the *whole* model, diluting the signal. **Inconclusive as measured:** a pooled-Euclidean-distance metric cannot cleanly isolate true near-lens backscatter from the ordinary population of soft/anti-aliasing Gaussians every 3DGS model accumulates. A per-camera-ray version (not pooled across all 300 overlapping camera positions) is the concrete fix, not yet built.

7.5 Initialization vs. optimization

Same underlying MAST3R-seeded reconstruction of PS117-29, before and after 3DGS's own photometric training: sonar r moves from **+0.067** (raw point cloud, right sign) to **−0.31** (after pure photometric training, no depth regularization). A real, if only 2-point, trajectory supporting the sharper claim: 3DGS does not fail to find usable initial geometry here—it finds a weakly-correct starting point and optimizes it away chasing photometric fit. A full dense trajectory on this exact config was not run in this pass (flagged as a direct extension using already-built tooling).

8. Three Follow-On Investigations

8.1 The PSNR/texture inversion — the project's strongest single finding

A collaborator's hypothesis: PSNR may be bounded by scene texture rather than reconstruction quality—PS117-29 (flat, monotone ice) scores a *higher* PSNR (38.12 dB) than PS117-39 (a richer, moving scene, 34–36 dB), backwards from what a “PSNR=quality” reading would predict.

Confirmed cleanly, then expanded from 2 stations/~500 frames to a pooled, systematic test across every existing trained model (no new training runs—inference-only extraction on already-trained checkpoints):

Scope	n	rho(gradient energy, PSNR)	p
PS117-39, pooled across 12 configs	3,298	−0.466	1.1e-177
PS117-29, pooled across 5 configs	1,632	−0.574	1.7e-143
Both stations pooled	4,930	−0.519	≈ 0 (machine precision)

13 of 17 individual configurations independently confirm the direction, 12 of those significant at $p < 0.001$. Recommended sentence for the paper: *“In under-ice imagery, PSNR is bounded by scene texture, not by geometric correctness—the least texturally-complex, hardest-to-reconstruct-in-3D frames systematically score the highest PSNR, because a smooth surface with little high-frequency content is trivial for photometric optimization to fit regardless of whether the underlying geometry is correct.”*

Caveat / limitation: True new capture *stations* (as opposed to new model configurations on the same 2 stations) were not achievable without new training—the third available station's dataset folder is empty and other candidate windows either lack trained models or are a confirmed-bad window from earlier in the project. Flagged rather than silently substituted.

8.2 Anisotropy with the orientation term

Covered in Section 7.3—built and run per this specific follow-up request. Result: the orientation correction resolves the puzzle (raw signal is a density/maturity confound) rather than rescuing the originally-hypothesized directional diagnostic.

8.3 Sonar weight sensitivity sweep

Varied sonar_loss_weight over 3 orders of magnitude (0.01 to 10.0), same rigorous held-out split as the coverage ablation, same base recipe:

weight	test r	test RMSE (m)
0.01	−0.501	0.395
0.03	−0.303	0.385
0.1 (chosen default)	−0.162 (best)	0.375 (best)
0.3	−0.427	0.390
1.0	−0.355	0.385
3.0	−0.332	0.381

10.0	−0.506	0.392
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A clean, well-behaved U-shaped curve: both extremes (0.01 and 10.0, a 1000× weight difference) land at nearly identical, worst values, with a single clear peak exactly at the project's already-chosen default of 0.1. This directly answers a reviewer who might suspect one lucky setting—0.1 is the empirically-verified optimum of a real sensitivity sweep, not an arbitrary choice, and the adaptive-gain scheme now has a concrete fixed-weight curve to be measured against.

Caveat / limitation: This sweep reuses the same block already known (from the coverage ablation) to be an anomalous, consistently wrong-signed region for PS117-39. Even the optimal weight stays negative—weight tuning clearly reduces the *magnitude* of error but does not flip this specific region's sign positive. A repeat on a favorable block (e.g. the one showing $r=+0.46$ to $+0.70$ in the coverage ablation) would show whether 0.1 is a genuinely global optimum or specific to difficult regions—not run in this pass.

9. Open Items and Recommended Next Steps

- Re-evaluate every config in the Section 5.1 rank-correlation table through a single, consistent contiguous-block holdout protocol (the fix already exists and is used elsewhere in this project) to produce a clean, unambiguous significance number for that specific statistic.
- Repeat the sonar weight sensitivity sweep (Section 8.3) on a favorable block to separate “optimal weight in general” from “optimal weight on a hard region.”
- Build the per-camera-ray (not pooled-Euclidean) version of the backscatter diagnostic (Section 7.4) to get a clean answer on whether 3DGS is representing the water column as geometry.
- Run a full dense-checkpoint trajectory on the vanilla, no-depth-reg MAST3R-seeded PS117-29 config (Section 7.5) to see the initialization-to-corruption dynamics, not just the two endpoints.
- The real-data spectral proof for Claim C (Section 3) needs either a simulated instrument (HoloOcean) or a real sequence flown at longer range than either existing tier.
- A third real capture station, if one becomes available with an existing or newly trained model, would let the texture-inversion finding (Section 8.1) claim new-station replication rather than new-configuration replication on the same two stations.

End of report. All underlying data, scripts, and per-finding markdown writeups referenced here are available in the project's results/ and scripts/ directories for full reproduction.